Technology-Driven Solutions for Tackling Domestic Violence

AN INDUSTRIAL ORIENTED MINI PROJECT REPORT

Submitted

in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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CERTIFICATE

This is to certify that the Industrial Oriented Mini project entitled “Technology-Driven Solutions for Tackling Domestic Violence” being submitted by B.Chandra Kanth(23B81A05CE), M.Tejaswi(23B81A05EA) in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to the CVR College of Engineering, is a record of bona fide work carried out by them under my guidance and supervision during the year 2025-2026.

The results embodied in this Industrial Oriented Mini project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Signature of the project guide	**Signature of the HOD**
Mr. M. Hanimi Reddy	**Dr. A Vani Vathsala**
Assistant Professor	**Department of CSE**

External Examiner

DECLARATION

I hereby declare that this project report titled "Technology-Driven Solutions for Tackling Domestic Violence" submitted to the Department of Computer Science and Engineering, CVR College of Engineering, is a record of original work carried out by me. The information and data presented in this report are true and authentic to the best of my knowledge. This project has not been submitted to any other university or institution for the award of any degree or diploma, nor has it been published at any time before.

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Date:

Place:

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We would like to express heart full thanks to Dr. A Vani Vathsala, Professor and Head of the Department of Computer Science and Engineering, who have always been grateful for us to complete our project.

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ABSTRACT

Domestic violence and online harassment are serious social issues that often remain unreported due to fear, social stigma, and lack of immediate support. Victims frequently hesitate to seek help during critical situations, which increases the risk and severity of harm. Existing systems such as helplines and social media reporting mechanisms rely on manual intervention and are often slow in providing timely assistance.

This project presents Protego, an AI-based safety system designed to provide real-time support through an intelligent chatbot interface. The system analyzes user messages using Natural Language Processing (NLP) and Machine Learning techniques to detect abusive, threatening, or distress-related language. Based on the analysis, the system classifies the input into different severity levels such as normal, distress, abusive, or severe.

According to the identified severity, the system generates appropriate responses, including emotional support, safety guidance, and emergency suggestions. In high-risk situations, the system recommends immediate actions such as contacting helplines or sharing the user's location using a panic feature. Additionally, the system maintains a database of detected incidents, including messages, severity levels, and timestamps for monitoring and analysis.

The proposed solution focuses on being lightweight, user-friendly, and efficient, making it suitable for real-time applications. By integrating AI-based detection with immediate guidance and emergency support, Protego aims to enhance user safety and provide a practical solution for tackling domestic violence and online harassment.

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CHAPTER 1

INTRODUCTION

Domestic violence and online harassment are serious social issues that often remain unreported due to fear, social stigma, and lack of immediate support. Many victims hesitate to seek help, especially during critical situations, which increases the risk and impact of harm. With the rapid growth of digital communication, there is a need for intelligent systems that can detect harmful interactions and provide timely assistance.

To address this problem, the proposed system Protego is an AI-based safety assistant designed to detect abusive, threatening, or distress-related language from user messages. The system uses Natural Language Processing (NLP) and Machine Learning techniques to analyze text input and classify it into different severity levels such as normal, distress, abusive, or severe. Based on this classification, the system provides appropriate responses including emotional support and safety guidance.

Unlike traditional systems that rely on manual reporting and delayed human intervention, Protego offers a real-time automated solution. It is implemented as a chatbot interface where users can communicate their situation in a simple and private way. In high-risk situations, the system suggests emergency actions such as contacting helplines or sharing the user’s location using a panic feature.

Additionally, the system stores detected incidents along with severity levels and timestamps in a database for monitoring and analysis. The design focuses on simplicity, accessibility, and real-time response, making it suitable as a lightweight safety support system.

In conclusion, Protego combines AI-based detection with real-time assistance to support individuals facing domestic violence and online abuse, helping improve awareness, response time, and user safety.

1.1 Motivation

The motivation behind this project arises from the increasing number of domestic violence and online harassment cases that often go unreported due to fear, social stigma, and lack of immediate support. Victims frequently hesitate to seek help, especially during critical situations, as they may not have quick access to trusted assistance or guidance.

Existing support systems such as helplines and reporting mechanisms rely heavily on manual intervention, which can lead to delayed responses. In many cases, victims require **instant support and guidance** to make safe decisions, but current systems do not provide real-time assistance.

With advancements in Artificial Intelligence and Natural Language Processing, it is possible to develop intelligent systems that can analyze user messages and identify distress or abusive situations automatically. This creates an opportunity to provide **immediate, private, and accessible support** without requiring direct human involvement at every step.

The motivation of this project is to build a system that not only detects abusive or distress-related language but also provides **practical safety guidance and emergency suggestions** to users. By integrating AI-based analysis with real-time interaction, the system aims to assist individuals in critical situations and improve awareness and response to domestic violence.

1.2 Problem Statement

Domestic violence and online harassment are major social issues that often remain unreported due to fear, social stigma, and lack of immediate support systems. Victims may hesitate to seek help because of privacy concerns, dependency, or unavailability of timely assistance. As a result, many situations escalate without early intervention.

Existing solutions such as helplines and reporting systems rely on manual processes and human intervention, which can lead to delays in response. Similarly, social media platforms depend on user-reported content for moderation, making the detection of harmful or abusive messages slow and inefficient. These systems do not provide real-time support or guidance to victims during critical situations.

There is a lack of an accessible and automated system that can analyze user messages, detect abusive or distress-related language, and provide immediate assistance. Current approaches focus mainly on detecting harmful content but do not actively support victims with safety guidance or emergency recommendations.

Therefore, there is a need to develop a **lightweight, real-time, and AI-based system** that can identify potential abuse, assess the severity of the situation, and guide users toward appropriate safety actions. Such a system can help bridge the gap between detection and support, enabling faster response and improving user safety.

1.3 Objectives

The main objectives of the proposed system are:

- To design an AI-based chatbot that can analyze user messages and detect abusive, threatening, or distress-related language.
- To classify user input into different severity levels such as **normal, distress, abusive, and severe** using machine learning techniques.
- To provide appropriate **safety guidance and supportive responses** based on the detected severity level.
- To suggest **emergency actions** such as contacting helplines or sharing location in high-risk situations.
- To implement a **real-time and user-friendly chat interface** for easy interaction.
- To store detected incidents, including messages, severity levels, and timestamps, in a database for monitoring and analysis.
- To develop a **lightweight and efficient system** that can be easily deployed without complex infrastructure.

1.4 Organization of the Project Report

This report is organized into six chapters, each describing different aspects of the Protego AI Safety System.

Chapter 1: Introduction

This chapter presents the background, problem statement, objectives, and scope of the project.

Chapter 2: Literature Review

This chapter discusses existing systems related to safety applications and highlights their limitations.

Chapter 3: System Analysis and Design

This chapter explains the proposed system, system architecture, and design diagrams such as use case, class, and sequence diagrams.

Chapter 4: Technology and Methodology

This chapter describes the technologies used and the methodology followed for developing the system.

Chapter 5: Implementation and Testing

This chapter provides details about system implementation, testing methods, and results obtained.

Chapter 6: Conclusion and Future Scope

This chapter summarizes the outcomes of the project and suggests future improvements.

CHAPTER 2

LITERATURE SURVEY

Author	Year	Findings	Limitations
Davidson et al.	2017	ML-based hate speech detection using text features	No safety guidance
Fortuna et al.	2020	Survey on hate speech detection techniques	No implementation for user support
Vidgen et al.	2021	Large datasets for abusive language detection	Focus only on classification
Jin et al.	2023	Studied temporal bias in abusive language detection models	Models degrade over time
Zhu et al.	2024	Proposed deep prompt multi-task network for better detection accuracy	Complex model, high computation
Multi-task LLM Studies	2025	Multi-task learning improves abusive language detection accuracy	Still lacks real-time assistance systems

2.1 Existing work

With the increasing use of social media and digital communication platforms, detecting abusive language and cyberbullying has become an important area of research. Many researchers have applied Natural Language Processing (NLP) and Machine Learning techniques to identify harmful or offensive content in text data.

Davidson et al. (2017) proposed machine learning-based approaches to classify hate speech and offensive language in social media. Their work focused on identifying abusive content using text features, achieving good classification accuracy. However, the system was limited to detection and did not provide any support or guidance to users.

Dinakar et al. (2012) developed methods for detecting cyberbullying in online conversations using NLP techniques. Their system was able to identify bullying patterns effectively, but it lacked real-time response capabilities and did not assist users in handling such situations.

Fortuna et al. (2020) presented a comprehensive survey on hate speech detection methods. Their work analyzed various techniques and datasets used in abusive language detection. However, the study focused only on analysis and did not propose a real-time solution for user support.

Vidgen et al. (2021) contributed by creating large datasets for abusive language detection and improving classification performance. While their work enhanced model training, it mainly focused on detection and did not address user safety or intervention mechanisms.

Recent studies such as Jin et al. (2023) and Zhu et al. (2024) explored advanced deep learning and transformer-based models for improving detection accuracy. These models showed better performance but are computationally complex and not suitable for lightweight real-time systems.

In addition to AI-based approaches, traditional support systems like helplines and reporting mechanisms are widely used to assist victims. Although these systems provide human support, they depend on manual intervention and often result in delayed responses.

2.2 Limitations of Existing Work

From the literature survey, it is observed that most existing systems for detecting abusive language and cyberbullying have several limitations.

Firstly, many systems focus only on **detecting harmful or offensive content** using machine learning or deep learning techniques. They do not provide any form of **support, guidance, or assistance to victims**, which is crucial in real-life situations.

Secondly, traditional support systems such as helplines and reporting mechanisms rely heavily on **manual intervention and human response**, which can lead to delays. Victims often require **immediate assistance**, especially during critical situations, but existing systems are not designed to provide real-time help.

Another limitation is that many advanced models, especially deep learning and transformer-based approaches, are **computationally complex and resource-intensive**. These models may not be suitable for lightweight applications or real-time deployment in practical environments.

Additionally, most existing solutions lack **integration of multiple functionalities** such as detection, severity classification, safety guidance, and emergency response within a single system. They typically address only one aspect of the problem.

Furthermore, current systems do not effectively consider **context or scenario-based understanding**, which is important for accurately assessing the seriousness of a situation.

Overall, there is a clear need for a system that not only detects abusive content but also provides **real-time safety guidance, emergency support, and a user-friendly interface**, which is addressed in the proposed system.

CHAPTER 3

SOFTWARE & HARDWARE SPECIFICATIONS

3.1 Software requirements

Component	**Description**
Programming Language	Python
Framework	FastAPI
Libraries	Natural Language Toolkit(NLTK)
Frontend Technologies	HTML, CSS, JavaScript
Database	SQLITE
Browser	Google Chrome/Microsoft Edge
Operating System	Windows / Linux

3.2 Hardware Requirements

Component	Specification
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	At least 5 GB free space
System Type	Laptop / Desktop

CHAPTER 4

PROPOSED SYSTEM DESIGN

4.0 Proposed methods

The proposed system follows a step-by-step process to detect abusive language and provide safety guidance.

Step 1: User Input

- User enters a message through the chatbot interface
- Message may contain normal, distress, or abusive content

Step 2: Text Preprocessing

- Convert text to lowercase
- Remove unnecessary characters
- Tokenization (split into words)
- Stop-word removal
- Lemmatization

Step 3: Feature Extraction

- Convert text into numerical format
- Use **TF-IDF vectorization**
- Extract important words/features

Step 4: Machine Learning Prediction

- Input features to trained ML model
- Model predicts severity level:
    1. Normal
    2. Distress
    3. Abusive
    4. Severe

Step 5: Risk Assessment

- Apply rule-based checks for critical keywords
- Identify scenario (home / public / general)
- Final severity level determined

Step 6: Response Generation

- Generate appropriate reply based on severity
- Provide:
    1. Emotional support
    2. Safety guidance
    3. Emergency suggestions

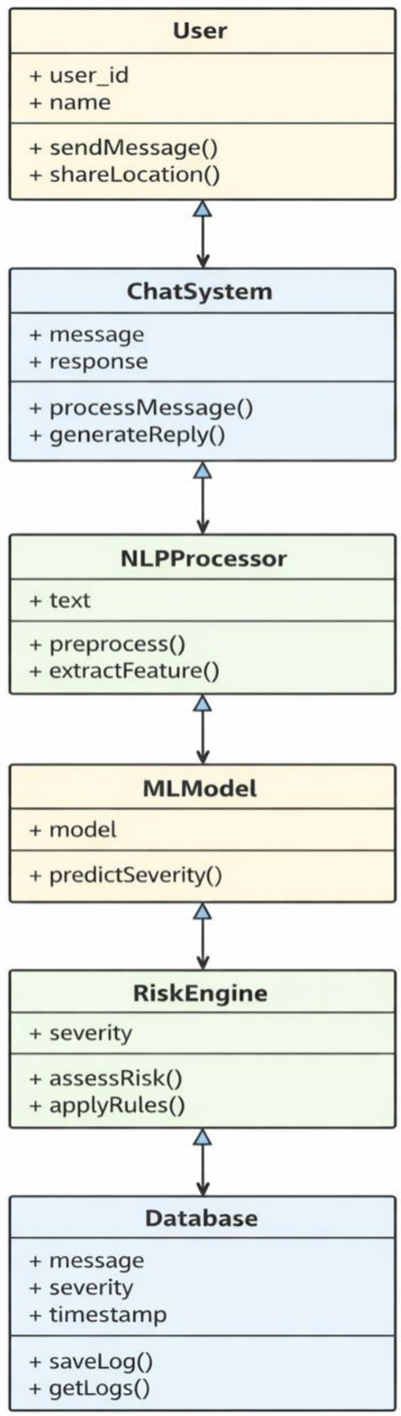
Step 7: Emergency Handling

- If severe case detected:
    1. Show helpline numbers
    2. Enable panic button
    3. Allow location sharing

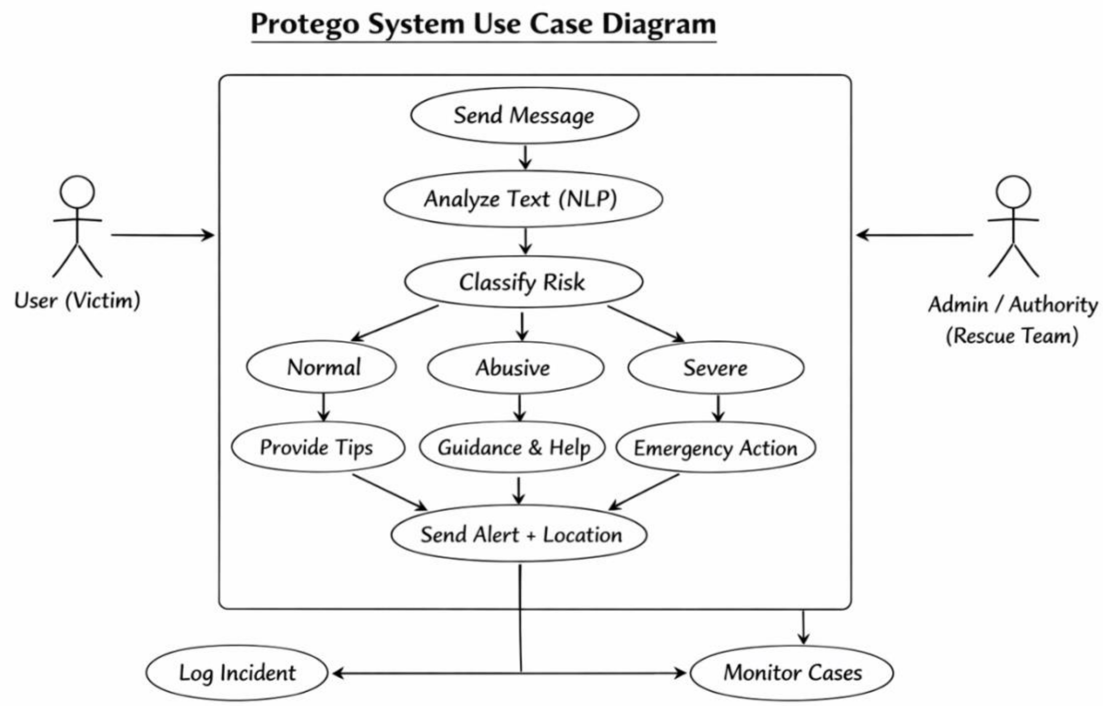
Step 8: Data Logging

- Store:
    1. User message
    2. Severity level
    3. Timestamp
    4. Location (if shared)
- Save in SQLite database

4.1 Class Diagram



4.2 Use Case Diagram



The Use Case Diagram represents the working of the Protego system, an AI-based chatbot designed to tackle domestic violence situations.

The primary actor is the User (victim) who interacts with the system by sending messages. The system processes these messages using Natural Language Processing (NLP) techniques to detect distress or abusive content.

After analysis, the system classifies the message into three categories:

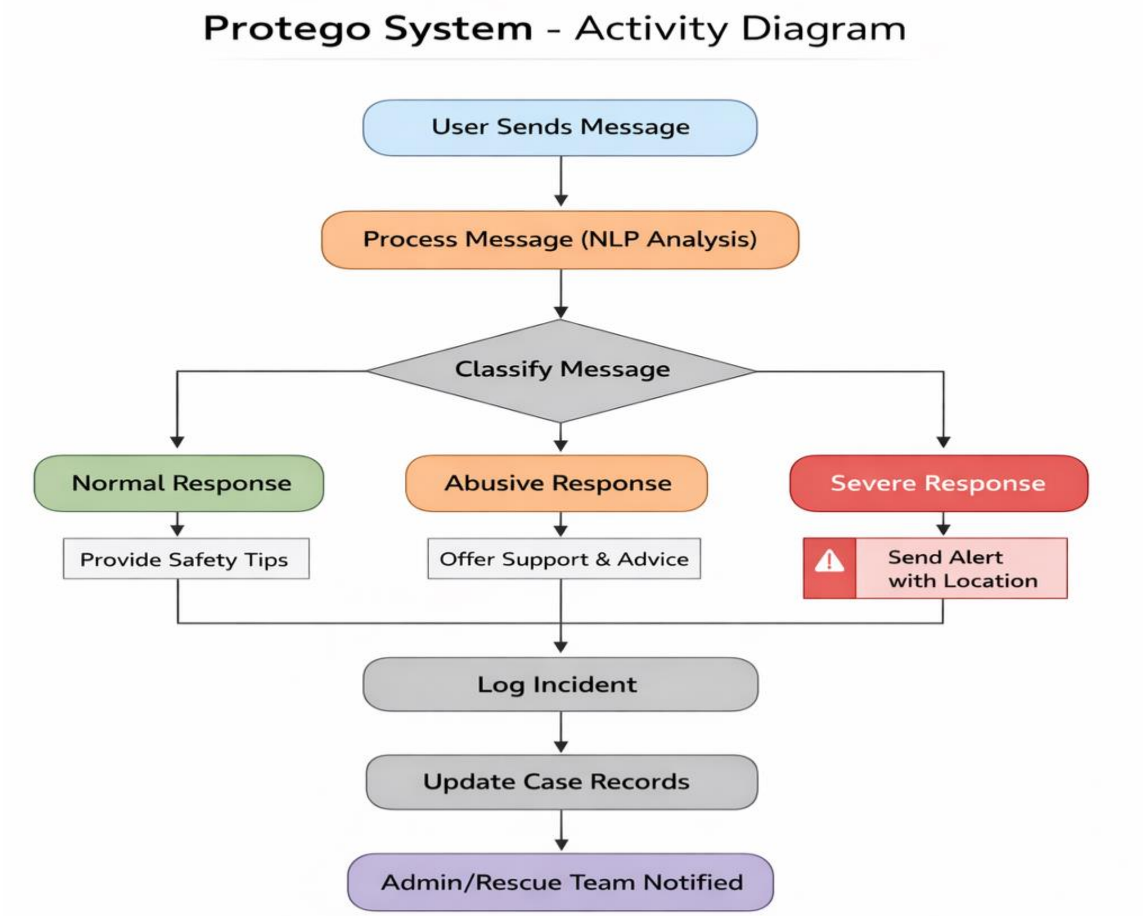
- Normal – provides general safety tips
- Abusive – offers guidance and support

- Severe – triggers emergency actions

  In severe cases, the system sends an alert along with the user's location to the Admin/Authority (rescue team) for immediate help.

  Additionally, all interactions are logged as incidents, and the admin can monitor cases for further action and support.

4.3 Activity diagram



The Activity Diagram illustrates the workflow of the Protego system in handling user messages related to domestic violence.

The process starts when the user sends a message to the chatbot. The system then performs NLP-based text processing to understand the content and detect distress or abusive language.

Next, the message is classified into three categories:

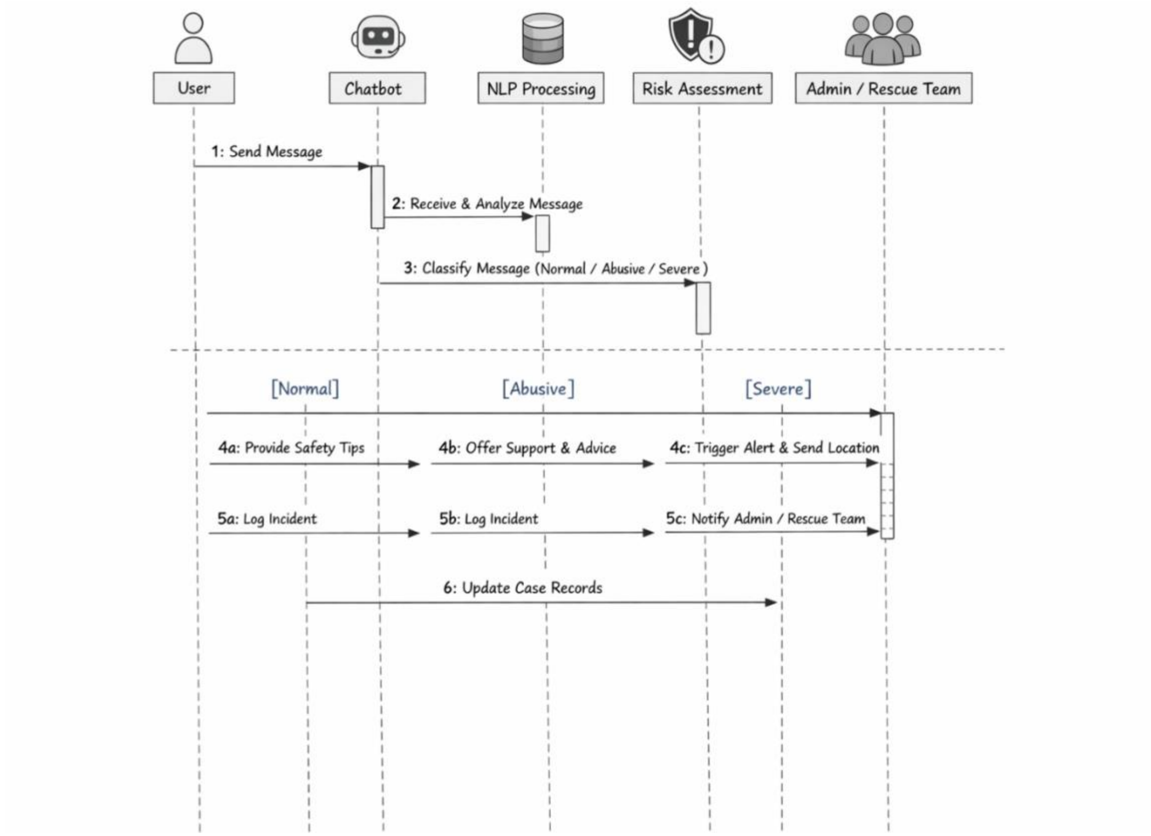
- Normal
- Abusive
- Severe

Based on the classification:

- For normal cases, the system provides safety tips.
- For abusive cases, it offers support and guidance.
- For severe cases, it triggers an emergency response by sending an alert along with the user's location.

After handling the response, the system logs the incident and updates case records. Finally, in critical situations, the admin or rescue team is notified for further action.

4.4 Sequence diagram



The Sequence Diagram represents the interaction between different components of the Protego system over time.

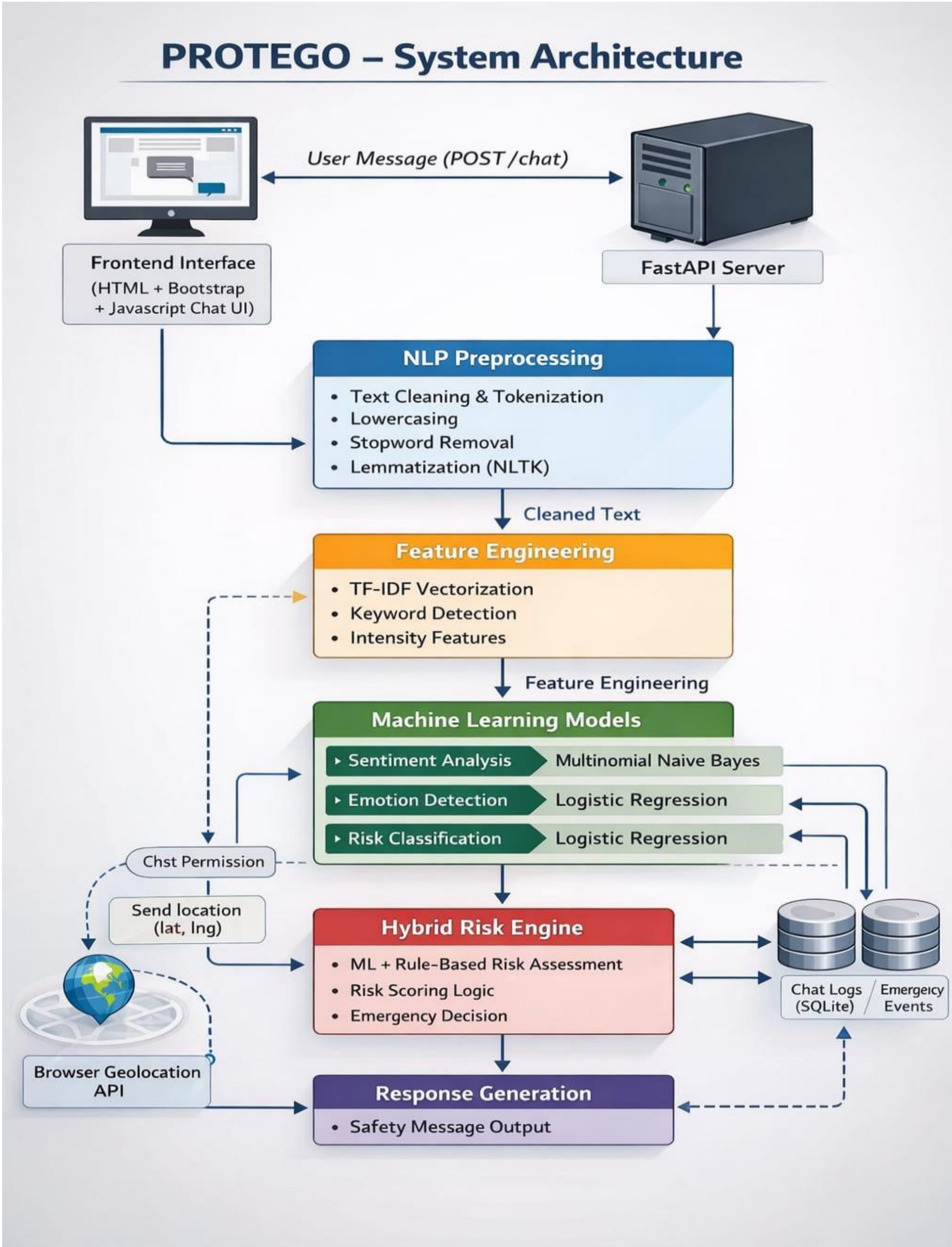
The process begins when the user sends a message to the chatbot. The chatbot receives the message and forwards it to the NLP processing module for analysis. The system then performs risk assessment by classifying the message into Normal, Abusive, or Severe categories.

Based on the classification:

- For normal cases, the chatbot provides safety tips to the user.
- For abusive cases, it offers support and guidance.
- For severe cases, the system triggers an emergency alert and sends the user's location to the admin/rescue team.

In all scenarios, the system logs the incident and updates the case records. In critical situations, the admin/rescue team is notified for immediate action.

4.5 System Architecture



The Protego system architecture represents the complete flow of data from user interaction to response generation.

The process begins at the frontend interface (HTML, Bootstrap, JavaScript), where the user sends a message. This message is sent to the FastAPI server, which handles backend processing.

The input text is then passed to the NLP preprocessing module, where tasks like text cleaning, tokenization, stopword removal, and lemmatization are performed to prepare the data.

Next, feature engineering is applied using techniques such as TF-IDF vectorization, keyword detection, and intensity analysis to extract meaningful features from the text.

These features are fed into machine learning models, including:

- Sentiment Analysis
- Emotion Detection
- Risk Classification

The outputs are then processed by the Hybrid Risk Engine, which combines machine learning results with rule-based logic to calculate risk scores and make emergency decisions.

If required, the system uses the browser geolocation API (with user permission) to fetch the user's location.

All interactions are stored in chat logs and emergency event databases (SQLite) for tracking and monitoring.

Finally, the system generates an appropriate response, such as safety advice or emergency alerts, and sends it back to the user.

4.6 Technology Description

The Protego system is developed using a combination of web technologies, backend frameworks, and machine learning techniques to provide real-time assistance.

💻 Frontend

The user interface is built using HTML, CSS (Bootstrap), and JavaScript, providing a simple and interactive chat-based environment for users to communicate with the system.

⚙️ Backend

The backend is implemented using FastAPI (Python), which handles user requests, processes data, and manages communication between different system components efficiently.

🧠 NLP & Libraries

The system uses Natural Language Processing (NLP) techniques with the help of the NLTK library. It performs:

- Text cleaning and tokenization
- Stopword removal
- Lemmatization

📊 Machine Learning

Machine learning models are used for:

- Sentiment Analysis
- Emotion Detection
- Risk Classification

Algorithms like Naive Bayes and Logistic Regression are used to classify messages into normal, abusive, or severe categories.

🗄️ Database

The system uses SQLite / Local Storage to store:

- Chat logs
- Emergency cases

- Incident records

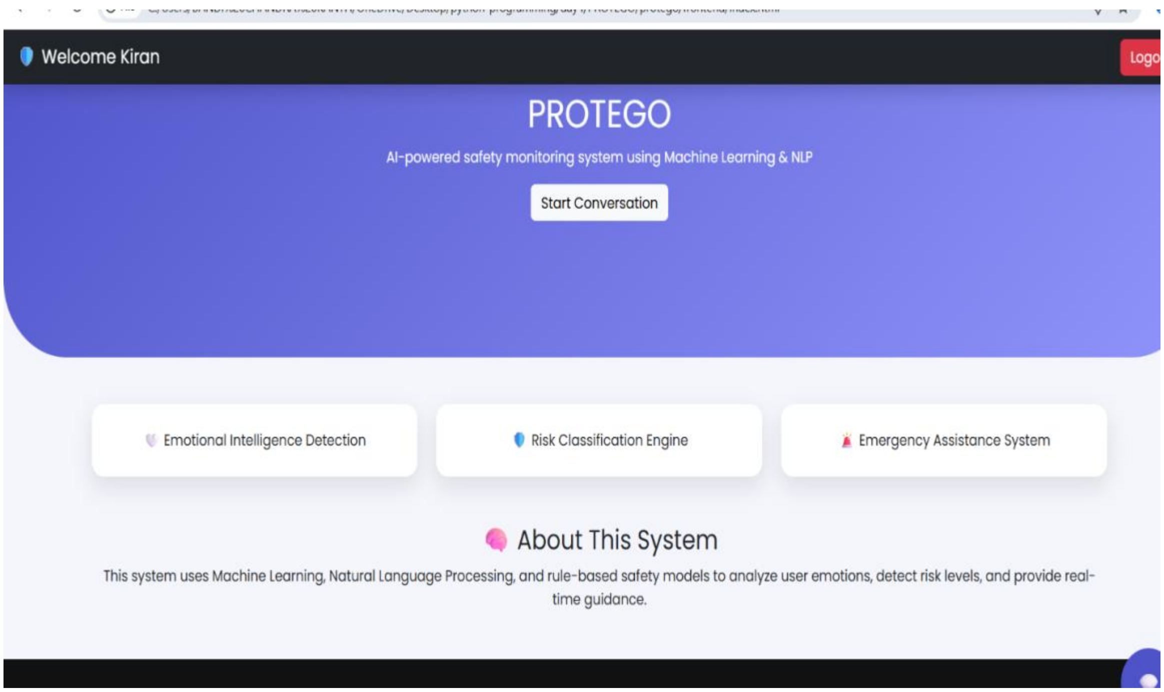
📍 Additional Technologies

- Browser Geolocation API – to capture user location in critical situations
- TF-IDF Vectorization – for feature extraction from text

CHAPTER 5

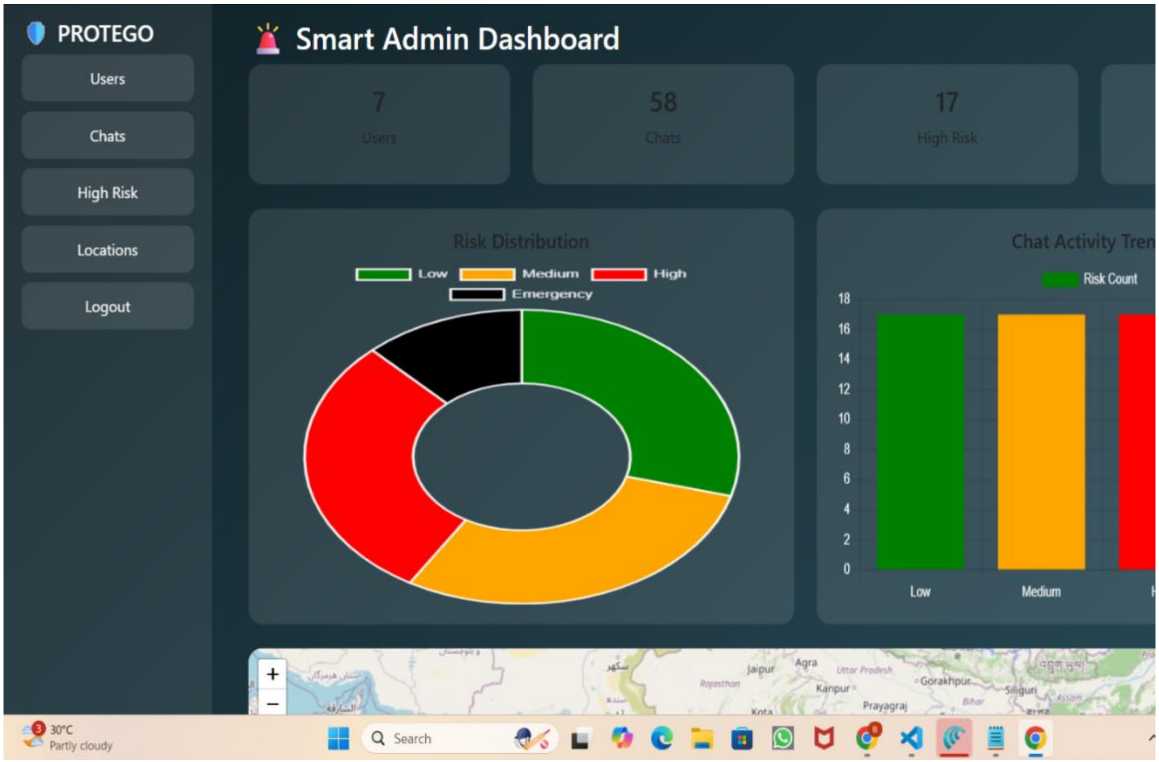
IMPLEMENTATION AND TESTING

5.1 Implementation



The above interface represents the main home page of the Protego AI Safety System, designed to provide a simple and user-friendly experience for users. It serves as the entry point to the application, allowing users to initiate interaction with the chatbot through the “Start Conversation” option. The interface highlights the core functionalities of the system, including Emotional Intelligence Detection, Risk Classification Engine, and Emergency Assistance System, which collectively work to ensure user safety.

Additionally, the page provides an overview of the system, explaining how it leverages Machine Learning, Natural Language Processing, and rule-based techniques to analyze user input, detect emotional states, and assess risk levels in real-time. The clean layout and intuitive design help users easily understand the system’s purpose and quickly access its features, making it effective for immediate support and guidance in critical situations.

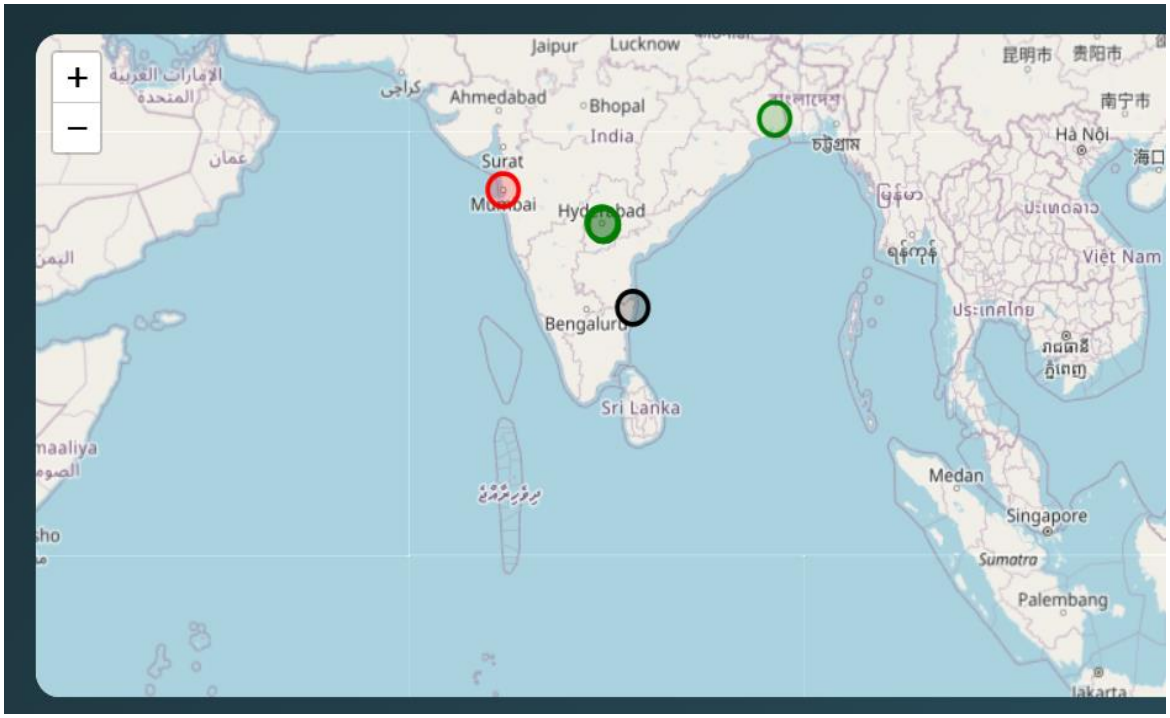


The Smart Admin Dashboard of the Protego AI Safety System provides a centralized platform for monitoring and managing user activity and safety-related events. It is designed with an intuitive and visually appealing interface that enables administrators to quickly analyze system performance and identify critical situations. The dashboard displays key metrics such as the total number of users, total chat interactions, and the number of high-risk cases detected by the system. These metrics help administrators gain a clear understanding of overall system usage and the frequency of emergency scenarios.

One of the core features of the dashboard is the visualization of risk distribution using graphical representations such as pie charts and bar graphs. These charts categorize user interactions into different risk levels, including low, medium, high, and emergency cases. This visual representation allows administrators to easily identify trends and patterns in user behavior, making it easier to detect an increase in critical situations and take timely action.

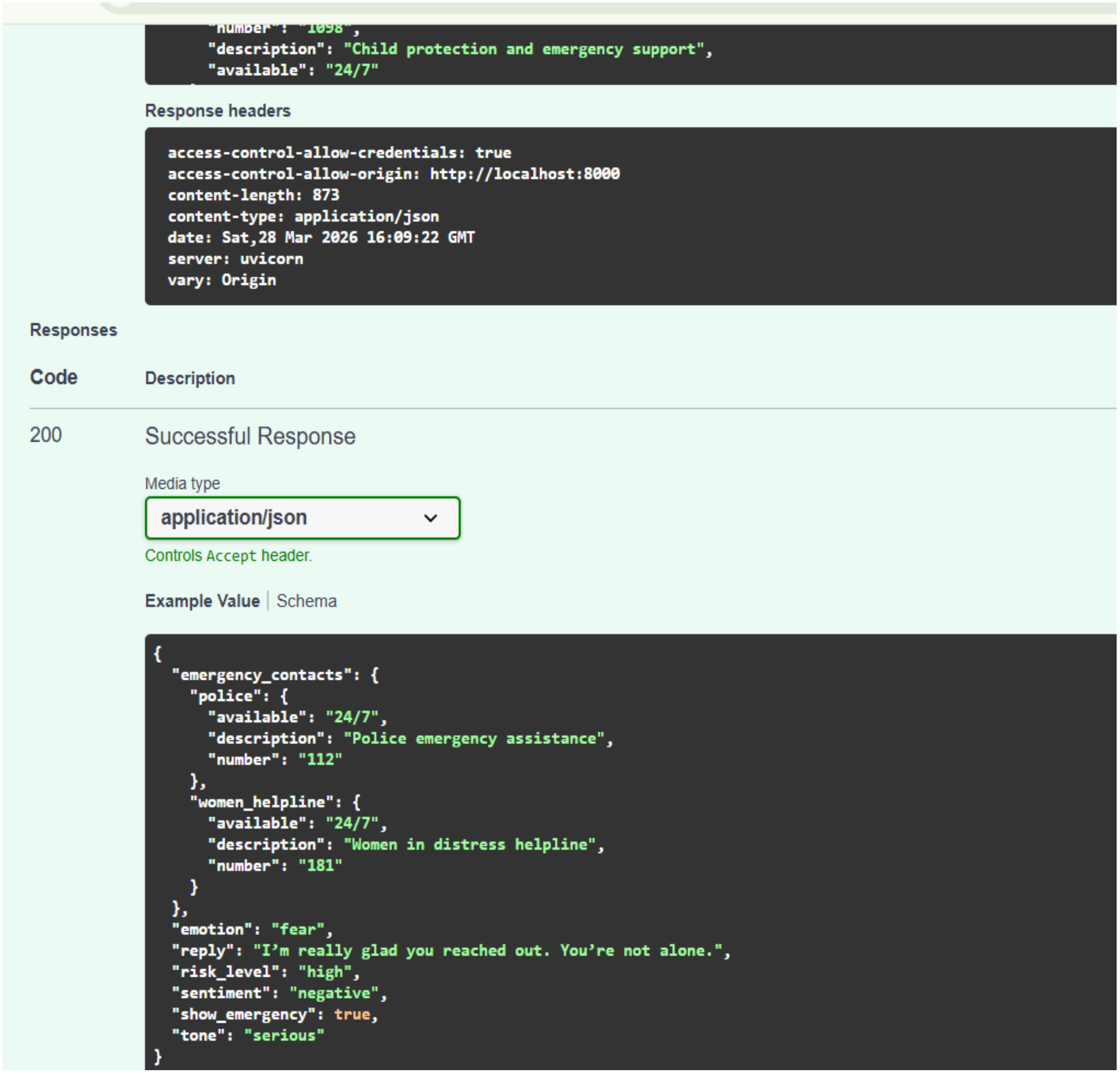
In addition to risk analysis, the dashboard also tracks chat activity trends, providing insights into how frequently users interact with the chatbot system. This helps in evaluating system engagement and effectiveness. The interface includes a navigation panel that allows administrators to access different sections such as users, chats, high-risk cases, and location data. This structured navigation ensures efficient management and quick access to relevant information.

Furthermore, the dashboard plays a crucial role in enhancing system responsiveness by enabling real-time monitoring of emergencies. When severe cases are detected, administrators can immediately view the associated data and initiate appropriate actions. Overall, the Smart Admin Dashboard acts as a powerful tool for decision-making, ensuring user safety by providing comprehensive insights, real-time monitoring, and effective management of risk-related data within the Protego AI system.



The above interface represents the location monitoring module of the Protego AI Safety System, which visualizes user locations on an interactive map. This module is used to track and display the geographical positions of users, especially in cases where emergency alerts are triggered. Different colored markers are used to indicate varying levels of risk, such as normal, medium, and severe conditions, enabling quick identification of critical situations. The map provides a real-time overview of user locations, allowing administrators or rescue teams to understand where incidents are occurring and respond effectively. By integrating geolocation services, the system ensures accurate location tracking when users activate the emergency feature. This enhances the efficiency of rescue operations by providing precise coordinates and visual representation on the map.

Overall, this module plays a crucial role in improving situational awareness and enabling faster response during emergencies, making the Protego system more effective in ensuring user safety and support.



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5.2 Testing

Testing is a crucial phase in the development of the Protego AI Safety System to ensure that all modules function correctly, efficiently, and reliably. The system was tested using various testing techniques to validate its performance, accuracy, and usability under different conditions. The primary goal of testing was to ensure that the chatbot accurately detects user risk levels, generates appropriate responses, and triggers emergency actions when required.

Functional Testing

Functional testing was performed to verify that each component of the system operates according to the specified requirements. The major functionalities tested include:

- User Authentication: The registration and login modules were tested to ensure that users can successfully create accounts and securely access the system. Invalid login attempts were also tested to verify error handling.
- Chatbot Interaction: The chatbot was tested with different user inputs to ensure that messages are correctly processed and appropriate responses are generated.
- Risk Classification: The machine learning model was tested using various text inputs representing normal, distress, and severe conditions to verify accurate classification.
- Emergency Alert System: The panic button and severe case detection were tested to confirm that location sharing and alert mechanisms (WhatsApp and email) are triggered correctly.
- Admin Dashboard: The dashboard was tested to ensure it correctly displays user statistics, risk levels, and system activity.

Performance Testing

Performance testing was conducted to evaluate the system's responsiveness and speed. The chatbot response time was measured to ensure that users receive real-time replies without noticeable delay. The system was tested with multiple inputs to verify its ability to handle continuous interaction efficiently. The results showed that the system provides quick responses, making it suitable for real-time safety applications.

Usability Testing

Usability testing was performed to ensure that the application is easy to use and understand. The user interface was evaluated for simplicity, clarity, and accessibility. Users were able to navigate through login, chatbot, and emergency features without confusion. The design was found to be intuitive, making it suitable for users in stressful or emergency situations.

Security Testing

Security testing was carried out to ensure that user data is protected. The authentication system was tested to prevent unauthorized access. Input validation was also checked to ensure that the system handles unexpected or invalid inputs safely. The use of secure login mechanisms ensures basic protection of user credentials.

Integration Testing

Integration testing was performed to verify that all modules of the system work together seamlessly. The interaction between frontend, backend, machine learning model, and database was tested. The system successfully integrates chatbot processing, risk detection, alert mechanisms, and data storage without errors.

5.3 Results

The Protego AI Safety System was successfully developed and tested to meet the objectives of providing real-time safety assistance using an AI-based chatbot. The system demonstrates effective performance in analyzing user inputs, classifying risk levels, and generating appropriate responses. The integration of Natural Language Processing and machine learning techniques enables the system to understand user messages and accurately categorize them into normal, distress, and severe conditions.

During testing, the chatbot consistently provided meaningful and context-aware responses based on the severity of the situation. For normal interactions, the system offered supportive and conversational replies, while for distress and severe cases, it generated more serious and guidance-oriented responses. This adaptive behavior enhances the system's ability to handle different user scenarios effectively.

One of the key outcomes of the system is its ability to trigger emergency actions in critical situations. When severe risk levels are detected, the system successfully activates alert mechanisms, including sharing the user's location and sending emergency messages via WhatsApp and email. This feature ensures quick response and improves the chances of providing timely assistance to users in danger.

The admin dashboard also plays a significant role in presenting system results. It provides a clear overview of user activity, total interactions, and distribution of risk levels. The use of visual elements such as charts and graphs allows easy analysis of system data, helping administrators monitor trends and identify critical cases efficiently.

In terms of performance, the system delivers fast response times, ensuring real-time interaction between the user and the chatbot. The user interface is simple, intuitive, and accessible, making it suitable for users even in stressful situations. The system also maintains reliable operation without major errors during testing.

Overall, the results indicate that the Protego AI Safety System is effective, reliable, and capable of providing intelligent safety support. It successfully achieves its goal of detecting risks, guiding users, and enabling emergency response, thereby contributing to improved personal safety and assistance.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

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The admin dashboard also plays a significant role in presenting system results. It provides a clear overview of user activity, total interactions, and distribution of risk levels. The use of visual elements such as charts and graphs allows easy analysis of system data, helping administrators monitor trends and identify critical cases efficiently.

In terms of performance, the system delivers fast response times, ensuring real-time interaction between the user and the chatbot. The user interface is simple, intuitive, and accessible, making it suitable for users even in stressful situations. The system also maintains reliable operation without major errors during testing.

Overall, the results indicate that the Protego AI Safety System is effective, reliable, and capable of providing intelligent safety support. It successfully achieves its goal of detecting risks, guiding users, and enabling emergency response, thereby contributing to improved personal safety and assistance.

6.2 Future Scope

The Protego AI Safety System is successfully designed and implemented as an intelligent web-based application aimed at addressing personal safety and domestic violence issues. The system integrates machine learning and Natural Language Processing techniques to analyze user inputs and classify them into different risk levels such as normal, distress, and severe.

The chatbot provides real-time, context-aware responses, offering guidance and support based on the detected situation. In critical cases, the system effectively triggers emergency alerts by sharing the user's location and sending notifications through WhatsApp and email, ensuring timely assistance. The inclusion of secure user authentication enhances data protection, while the admin dashboard provides a comprehensive view of user activity and risk distribution for efficient monitoring.

The system demonstrates reliable performance, fast response time, and user-friendly interaction, making it suitable for real-time safety applications. Overall, Protego achieves its objective of providing an intelligent, responsive, and accessible safety solution, contributing to improved support for individuals in vulnerable situations.

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